

Alexe James
Individual assignment

Q1 Formulating LPP

Choco = 3 cups of flour & 2 cups of sugar
vanilla = 5 cups " & 1 cup "

Total cups of flour = 390

" Sugar = 130

cost to produce choco = \$5

" vanilla = \$4

	flour (x)	sugar (x)	cost (\$)
choco (x_1)	3	2	5
vanilla (x_2)	5	1	4
Total cups	390	130	

$$\text{Min } Z = 5x_1 + 4x_2$$

Subject to

$$3x_1 + 5x_2 \leq 390$$

$$2x_1 + x_2 \leq 130$$

$$x_1, x_2 \geq 0$$

$$\begin{aligned} 0 - 1(1) &= 0 \\ 0 - 0(1) &= 0 \\ 1 - 0(0) &= 0 \\ 1 - 0(0) &= 0 \end{aligned}$$

use big M

$$\text{minimize } P = x_1 - x_2 + 3x_3$$

s.t

$$x_1 + x_2 \leq 20$$

$$x_1 + x_3 = 5$$

$$x_2 + x_3 \geq 10$$

$$x_1, x_2, x_3 \geq 0$$

Soln

$$-1(x_1 - x_2 + 3x_3)$$

$$\text{max } P = -x_1 - x_2 + 3x_3 + 0s_1 + 0s_2 - Ma_1 - Ma_2$$

st

$$x_1 + x_2 + s_1 = 20$$

$$x_1 + x_3 + a_1 = 5$$

$$x_2 + x_3 - s_2 + a_2 = 10$$

$$x_1, x_2, x_3, s_1, s_2 \geq 0$$

entering

B		x_1	x_2	x_3	s_1	s_2	a_1	a_2		
	c_j	-1	1	-3	0	0	-M	-M	b	ratio
s_1	0	1	1	0	1	0	0	0	20	20
a_1	-M	1	0	1	0	0	1	0	5	-
a_2	-M	0	1	1	0	-1	0	1	10	10
	z_j	-M	-M	-2M	0	M	-M	-M		
leaving	$z_j - c_j$	1-M	-1-M	3-2M	0	-M	0	0		

entering

B		x_1	x_2	x_3	s_1	s_2	a_1	
	c_j	-1	1	-3	0	0	-M	b
s_1	0	1	0	-1	1	1	0	10
a_1	-M	1	0	1	0	0	1	5
x_2	1	0	1	1	0	-1	0	10
	z_j	-M	1	-M+1	0	-1	-M	

$$s_1 = s_1 \leftarrow \text{new } x_2$$

$$a_1 =$$

$$z_j - c_j \quad 1-M \quad 0 \quad 4-M \quad 0 \quad -1 \quad 0$$

B		x_1	x_2	x_3	s_1	s_2	
s_1	c_j	-1	1	-3	0	0	b
s_2	0	0	0	-2	1	1	5
x_1	-1	1	0	1	0	0	5
x_2	1	0	1	0	1	0	15
	z_j	-1	1	-1	1	0	10
	$z_j - c_j$	0	0	2	1	0	

$$\text{Max } Z = 10$$

$$\therefore \text{min } P = -10$$

$$\text{at } x_1 = 5$$

$$x_2 = 15$$

$$x_3 = 0$$